Year 11 Higher

REVISION PACK

Algebra 7

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Algebra 7 · Formulae (Lessons 2.1, 2.2)

Q	Working	Answer
substitution		
1.1	$s = 2 \times 3 + \frac{1}{2} \times (-4) \times 3^2$ $= 6 - 18$ $= -12$	$s = -12$
1.2	$s = 6 \times 2 + \frac{1}{2} \times (-5) \times 2^2$ $= 12 - 10$ $= 2$	$s = 2$
1.3	$s = 4 \times 5 + \frac{1}{2} \times (-1.5) \times 5^2$ $= 20 - 18.75$ $= 1.25$	$s = 1.25$
1.4	$s = 1 \times 4 + \frac{1}{2} \times (3) \times 4^2$ $= 4 + 24$ $= 28$	$s = 28$
1.5	$s = 10 \times 2 + \frac{1}{2} \times (-2.5) \times 2^2$ $= 20 - 5$ $= 15$	$s = 15$
1.6	$s = 3 \times 6 + \frac{1}{2} \times (-2) \times 6^2$ $= 18 - 36$ $= -18$	$s = -18$
rearranging formulae		

1.7	**(a)** $y-7=4x$ $x=\dfrac{y-7}{4}$ **(b)** $r^2=\dfrac{A}{3}$ $r=\sqrt{\dfrac{A}{3}}$ **(c)** $5x-cx=2-b$ $x(5-c)=2-b$ $x=\dfrac{2-b}{5-c}$	**(a)** $x=\dfrac{y-7}{4}$ **(b)** $r=\sqrt{\dfrac{A}{3}}$ **(c)** $x=\dfrac{2-b}{5-c}$
1.8	**(a)** $y+9=2x$ $x=\dfrac{y+9}{2}$ **(b)** $r^2=\dfrac{V}{\pi h}$ $r=\sqrt{\dfrac{V}{\pi h}}$ **(c)** $ax-4x=b+3$ $x(a-4)=b+3$ $x=\dfrac{b+3}{a-4}$	**(a)** $x=\dfrac{y+9}{2}$ **(b)** $r=\sqrt{\dfrac{V}{\pi h}}$ **(c)** $x=\dfrac{b+3}{a-4}$
1.9	**(a)** $y-1=6x$ $x=\dfrac{y-1}{6}$ **(b)** $v^2=\dfrac{2K}{m}$ $v=\sqrt{\dfrac{2K}{m}}$ **(c)** $7x-qx=3-p$ $x(7-q)=3-p$ $x=\dfrac{3-p}{7-q}$	**(a)** $x=\dfrac{y-1}{6}$ **(b)** $v=\sqrt{\dfrac{2K}{m}}$ **(c)** $x=\dfrac{3-p}{7-q}$

1.10	**(a)** $y+2=7x$ $x=\dfrac{y+2}{7}$ **(b)** $t^2=\dfrac{P}{5}$ $t=\sqrt{\dfrac{P}{5}}$ **(c)** $px-2x=q-5$ $x(p-2)=q-5$ $x=\dfrac{q-5}{p-2}$	**(a)** $x=\dfrac{y+2}{7}$ **(b)** $t=\sqrt{\dfrac{P}{5}}$ **(c)** $x=\dfrac{q-5}{p-2}$
1.11	**(a)** $y-8=3x$ $x=\dfrac{y-8}{3}$ **(b)** $r^2=\dfrac{S}{4\pi}$ $r=\sqrt{\dfrac{S}{4\pi}}$ **(c)** $4x-bx=6+a$ $x(4-b)=6+a$ $x=\dfrac{6+a}{4-b}$	**(a)** $x=\dfrac{y-8}{3}$ **(b)** $r=\sqrt{\dfrac{S}{4\pi}}$ **(c)** $x=\dfrac{6+a}{4-b}$
1.12	**(a)** $y+4=9x$ $x=\dfrac{y+4}{9}$ **(b)** $x^2=\dfrac{A}{6}$ $x=\sqrt{\dfrac{A}{6}}$ **(c)** $nx-2x=m+7$ $x(n-2)=m+7$ $x=\dfrac{m+7}{n-2}$	**(a)** $x=\dfrac{y+4}{9}$ **(b)** $x=\sqrt{\dfrac{A}{6}}$ **(c)** $x=\dfrac{m+7}{n-2}$
changing the subject		
1.13	$u^2=v^2-2as$ $u=\sqrt{v^2-2as}$	$u=\sqrt{v^2-2as}$

1.14	$2w = P - 2l$ $w = \dfrac{P - 2l}{2}$	$w = \dfrac{P - 2l}{2}$
1.15	$2A = (a + b)h$ $h = \dfrac{2A}{a + b}$	$h = \dfrac{2A}{a + b}$
1.16	$y - c = mx$ $m = \dfrac{y - c}{x}$	$m = \dfrac{y - c}{x}$
1.17	$3V = \pi r^2 h$ $h = \dfrac{3V}{\pi r^2}$	$h = \dfrac{3V}{\pi r^2}$
1.18	$3b^2 = T - 2a$ $b^2 = \dfrac{T - 2a}{3}$ $b = \sqrt{\dfrac{T - 2a}{3}}$	$b = \sqrt{\dfrac{T - 2a}{3}}$